

Measurement of National Income

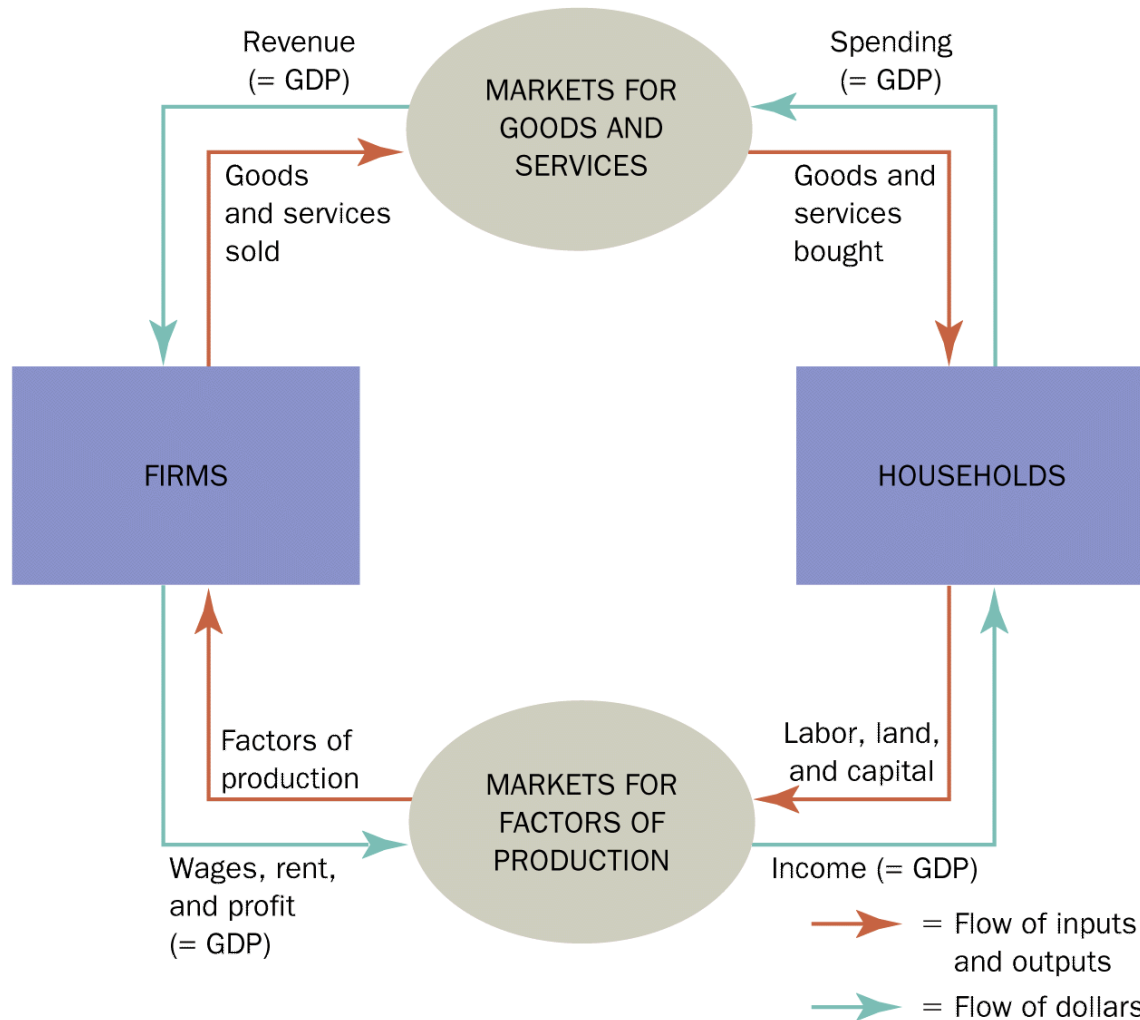
06 October 2025

Introduction

- How do we assess economic health of a nation?
 - Let's begin with a household
 - Income?
- Why is income important?
 - Standard of living
 - Likewise, nations with high income can afford better education, health care and other amenities for their citizens
- The gross domestic product (GDP) is a measure of
 - Total income of everyone in the economy
 - Total expenditure on goods and services
- For an economy as a whole, its income must equal expenditure
 - Every transaction has a buyer and a seller
 - Example
- GDP is believed to be a good measure of a society's *economic* well-being
 - The most important measure to assess *overall performance* of an economy
 - If we ask an economic historian, what happened to the economy during the Great Depression or after the WW-II, the answer will include changes in GDP

The circular flow diagram

Note: actual economy is much more complex (more agents, more instruments – taxes, savings, etc.,)



Defining GDP

- GDP is the market value of all final goods and services produced within a country in a given period of time
 - Market value
 - Value: comparing apples and oranges
 - Market price (price: willingness to pay; value)
 - Of all final goods and services
 - Goods (food, clothing), services (haircut, health), housing
 - GDP includes the market value of housing services; for rental housing it is the value of rent whereas for owner occupied houses, the value is imputed
 - Final goods and services (double counting?)
 - Many goods are processed several times. The value of intermediate goods is already included in final goods
 - Produced within a country
 - Only the goods and services currently produced are counted. Selling a used car is not counted
 - Production within a country
 - An IIT graduate working in the US
 - In a specific period of time
 - Usually annual frequency, quarterly GDP

Approaches to GDP Measurement

- As shown in the circular flow diagram, GDP may be estimated through different techniques
- Expenditure approach
 - $GDP = C + I + G + NX$, $NX = \text{Exports} - \text{Imports}$
- Income approach
 - the sum of income (factor rewards) for the factors of production
- Production (value added) approach
 - Consider a farmer who produces wheat. It costs him Rs 0 to produce wheat. The farmer sells the wheat to a flour factory for Rs 5. The factory sells it to a bakery shop for Rs 10. Finally, a consumer buys it for Rs 15
 - $GDP = \text{sum of the value added in the three stages}$
 - $GDP = (5-0) + (10-5) + (15-10) = 15$
- In practice, NAS are compiled from a mix of the three approaches, using a wide array of
 - data sources, and
 - methods

Equivalence of the approaches (3 firms in the economy)

Total spending on domestically produced final goods and services = \$21,500

	American Ore, Inc.	American Steel, Inc.	American Motors, Inc.	Total factor income
Value of sales	\$4,200 (ore)	\$9,000 (steel)	\$21,500 (car)	
Intermediate goods	0	4,200 (iron ore)	9,000 (steel)	
Wages	2,000	3,700	10,000	\$15,700
Interest payments	1,000	600	1,000	2,600
Rent	200	300	500	1,000
Profit	1,000	200	1,000	2,200
Total expenditure by firm	4,200	9,000	21,500	
Value added per firm	4,200	4,800	12,500	
=				
Value of sales – cost of intermediate goods				

Total payments to factors = \$21,500

Sum of value added = \$21,500

Treatment of investment and transfers

- Investment in macroeconomics
 - Real vs financial investment
 - Gross vs net investment
 - Net investment is indicative of the increase in a nation's *capital stock*
 - Along with consumption goods and services, gross investment is included in GDP
 - All investment goods – plant and machinery, housing – produced in an economy are included in GDP
- Government expenditure
 - Government purchases and transfer payments
 - Transfer payments are payments that are not made in exchange of goods and services supplied
 - Examples: scholarships, social security benefits, in-kind transfers (free foodgrains)
 - Transfer payments are excluded but factor payments are included

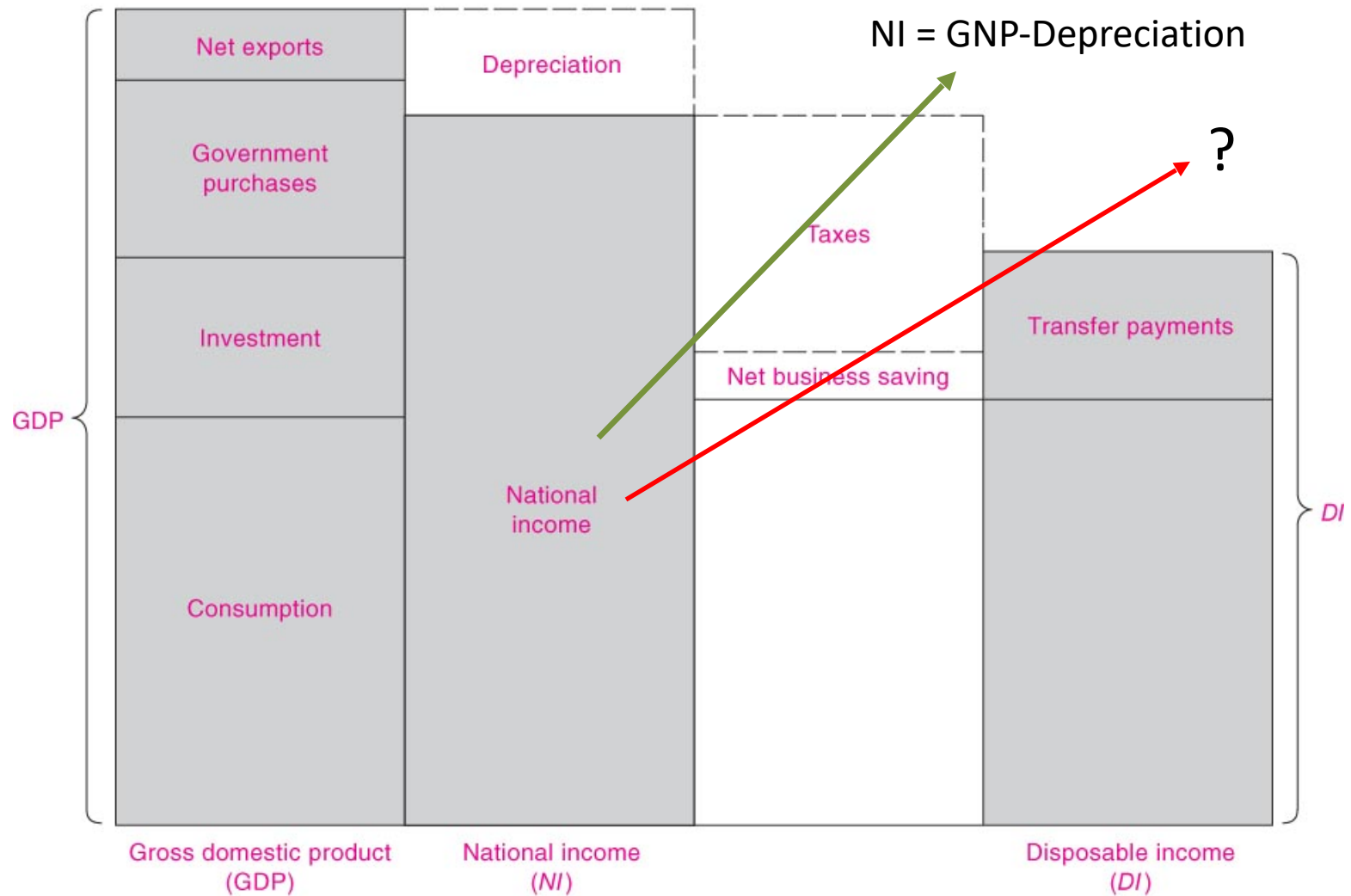
The GDP family

- Gross National Product (GNP)
 - Total income earned by the citizens
 - $\text{GNP} = \text{GDP} + \text{Net factor income from abroad}$
- Gross and Net
 - Net means net of depreciation, depreciation is wear and tear of the capital stock in an economy
 - Thus, $\text{Net} = \text{Gross} - \text{depreciation}$
 - NDP and NNP
- GDP at factor cost and GDP at market price
 - $\text{GDP at MP} = \text{GDP at FC} + (\text{Indirect taxes} - \text{subsidies})$

Factor cost and market price

- In the presence of distortions (including government interventions such as indirect taxes or subsidies), market value does not equal value added
- Cigarettes (taxes)
 - If the market price is Rs 20, tax is Rs 2
 - GDP at MP = Rs 20, GDP at FC = 18

From GDP to National Income



India's GDP

- Any idea about India's GDP
 - Rs 20,351,013 Crores in 2019-20 (NAS 2021)
- Per capita GDP is the GDP per population
 - Rs 151,760 in 2019-20 (NAS 2021)
- NNP is generally known as the national income and per capita NNP as per capita income (PCI)
- Statistical discrepancy
 - GDP estimates from different approaches may not be identical
 - Data sources are different and are not perfect

Quantity	Value
GDP	20,351,213
NDP	18,187,414
<i>C</i>	<i>12,309,019</i>
<i>G</i>	<i>22,85,016</i>
<i>I (GCF; public + private)</i>	<i>65,51,251</i>
<i>X</i>	<i>37,50,567</i>
<i>M</i>	<i>42,65,040</i>
NFIA	-1,93,114
GNP or GNI	20,157,899
NNP or NI	17,994,301
GDP pc	151,760
PCI	134,186
<i>Savings (Gross)</i>	<i>6,385,981</i>

Components of GDP

- Consumption (C)
 - Expenditure by households on goods and services
 - **Goods**: durable (refrigerator) and nondurable (food)
 - **Services** include educational and health services
- **Investment (I)**
 - **Goods** that are in turn **used to produce goods and services**
 - Capital equipment, (changes in) inventory, and new housing
- **Government Expenditure (G)**
 - **Spending by central, state and local governments**
 - **Salaries, expenditure on public works**
 - As discussed, **transfer payments would not be added as** they are “not made in exchange for a currently produced good or service”
- Net Exports (NX)
 - $NX = \text{Exports} - \text{Imports}$; needed to account for the fact that the economies undertake trade with the rest of world; imports are subtracted as GDP as the components of GDP (e.g., laptop) may include imports

Components of GDP

$$\text{GDP} = C + I + G + \text{NX}$$

Component	USA (2009, billion USD)		India (2009, Crore Rupees)	
	Amount	Share (%)	Amount	Share (%)
GDP	14,259		6,457,352	
C	10,093	71	3,708,136	57
I	1,623	11	2,363,670	36
G	2,933	21	774,272	12
NX	-390	-3	-346,902	-5

India's GDP

Sectoral composition in 2019 (NAS 2021)

2019 ▼	Item	▼
18.4	agriculture, forestry and fishing	1
1.9	mining and quarrying	2
14.7	manufacturing	3
2.6	electricity, gas, water supply & other utility services	4
7.4	construction	5
12.6	trade, repair, hotels and restaurants	6
6.3	transport, storage, communication & services related to broadcasting	7
5.7	financial services	8
15.5	real estate, ownership of dwelling & professional services	9
6.3	public administration and defence	1
8.6	other services	1
100.0	TOTAL GVA at basic prices	1

Real and nominal variables

- **Nominal** variable
 - A variable measured over time **without adjustment for the changing value of currency**
- **Real** variable
 - A variable **adjusted for the changing value of currency**
- Quite a few macroeconomic statistics are reported in both nominal and real terms
- The distinction between the two is crucial in macroeconomics
- To track **inter-temporal changes**, we **use real variables**
 - This helps us eliminate the effect of price changes (example in the next slide)

Real and Nominal GDP

- If the GDP rises in the next year, it could be due to two reasons
 - Economy's production has increased
 - Prices of the goods and services being produced have increased
- Our interest lies in measuring the effect that is not due to changes in prices. Hence, we use the real GDP
 - Real GDP is the value of goods and services produced this year and valued at the prices prevailing in the base year (last year, for example)
 - Nominal GDP, on the other hand, is the GDP valued at the current year market prices
- GDP deflator
 - Is the ratio (in %) of nominal to real GDP
 - Is a measure of the price level

Real and Nominal GDP

Prices and Quantities

Year	Price of Hot Dogs	Quantity of Hot Dogs	Price of Hamburgers	Quantity of Hamburgers
2001	\$1	100	\$2	50
2002	2	150	3	100
2003	3	200	4	150

Year Calculating Nominal GDP

2001	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \200
2002	$(\$2 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$3 \text{ per hamburger} \times 100 \text{ hamburgers}) = \600
2003	$(\$3 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$4 \text{ per hamburger} \times 150 \text{ hamburgers}) = \$1,200$

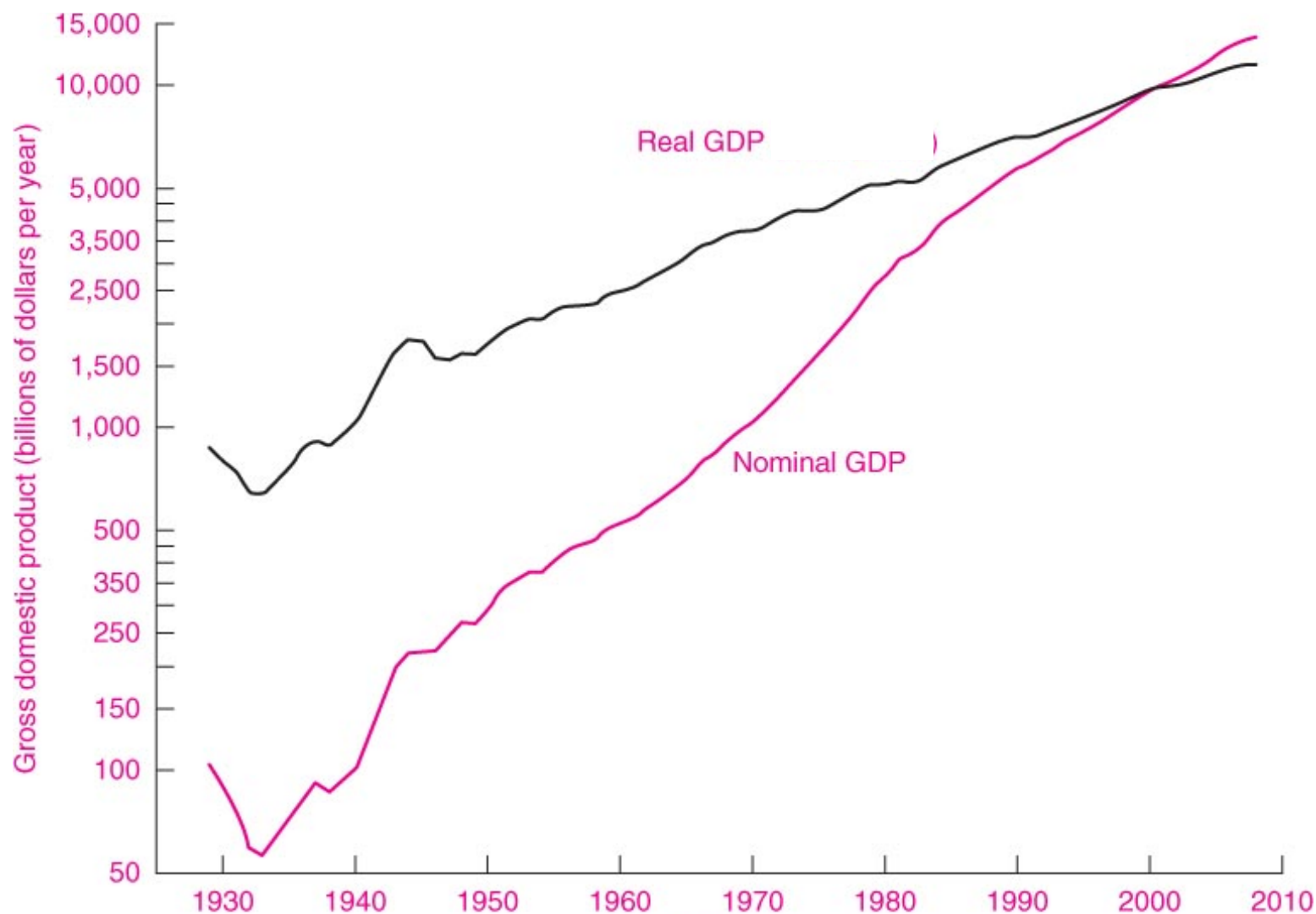
Year Calculating Real GDP (base year 2001)

2001	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \200
2002	$(\$1 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 100 \text{ hamburgers}) = \350
2003	$(\$1 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 150 \text{ hamburgers}) = \500

Year Calculating the GDP Deflator

2001	$(\$200/\$200) \times 100 = 100$
2002	$(\$600/\$350) \times 100 = 171$
2003	$(\$1,200/\$500) \times 100 = 240$

Real and nominal GDP: which one is growing fast?



India's GDP: more numbers

Source: NAS 2021

2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Item
3	4	5	6	7	8	9	10	11	2
1220	1235	1251	1267	1283	1299	1314	1327	1341	Population (in million)
									At current prices
71609	80518	89796	98405	107341	118489	130061	142328	151760	Per Capita GDP(₹)
70980	79573	88678	97242	106096	117131	128655	140804	150320	Per Capita GNI (₹)
63462	70983	79118	86647	94797	104880	115224	125883	134186	Per Capita NNI (₹)
73479	82408	91843	100439	109315	120052	131743	144524	154349	Per Capita GNDI(₹)
40250	45461	51764	57201	63339	70258	76379	84567	91790	Per Capita PFCE(₹)
									At constant (2011-12)
71609	74599	78348	83091	88616	94751	100035	105526	108645	Per Capita GDP (₹)
70980	73722	77370	82107	87565	93638	98925	104377	107589	Per Capita GNI (₹)
63462	65538	68572	72805	77659	83003	87586	92241	94566	Per Capita NNI(₹)
40250	41936	44423	46667	49738	53120	55789	59415	62056	Per Capita PFCE(₹)
									Percentage change over previous year at constant (2011-12) prices
	4.2	5.0	6.1	6.6	6.9	5.6	5.5	3.0	Per Capita GDP
	3.9	4.9	6.1	6.6	6.9	5.6	5.5	3.1	Per Capita GNI
	3.3	4.6	6.2	6.7	6.9	5.5	5.3	2.5	Per Capita NNI
	4.2	5.9	5.1	6.6	6.8	5.0	6.5	4.4	Per Capita PFCE

GDP and well being

- Is GDP a perfect measure?
 - Transactions which take place at the market are easily recorded whereas non-market transactions are not
 - Barter and corporate transactions
 - Vegetables bought from market and grown in kitchen garden
 - A chef preparing meals for the restaurant and for home (similarly, the marriage of a man with his maid may result in reduced GDP)
 - Services rendered by women in houses (and day care fee?)
 - Environment
 - What if there are no government regulations and producers exploit the natural resources
 - Externalities (cost to the society not counted)
 - How do we value fresh air?
 - Ronald Wright (historian): “If civilization is to survive, it must live on the interest, not the capital, of nature”
 - Distribution of income: inequality
 - All the things that contribute to a good quality of life not accounted for. Example, leisure

GDP and the underground economy

- GDP includes all items produced in the economy and sold legally in markets
 - GDP excludes the items produced and sold illicitly; such items do not enter the marketplace
 - Examples: illegal drugs, criminal activities, gambling, hunting, etc.
- In the US, the size of the underground economy is estimated to be between 6 and 20 per cent
- Bolivia and Zimbabwe: nearly two-third of the output
- India: 17-25 per cent (Chaudhuri et al. 2005)
- It is generally agreed that the size of shadow economy is larger in developing countries: their average was about 40 % compared to 16 for the West

GDP and the third world countries

- The previous discussion has highlighted some of the difficulties in accurately estimating the GDP for the third world countries
 - For example, contribution of agriculture sector, large presence of informal sector, value of working from home
 - Also, the lack of resources and skilled manpower
- What impact would these have on the GDP of the third world countries?
 - Under-estimation for countries like India?
 - *“Africa’s Statistical Tragedy”*
- It is worth noting that GDP was developed in the aftermath of the Second World War and subsequently became popular so much so that it became an “aspirational goal” for nations
 - It is a measure inherently suited more to the industrialized countries

Price indices

Defining a price index

- A price index (P) measures the average level of prices
- Inflation (π) is a rise in the general price level (P). The inflation rate is measured as follows
- Generally, three price indices are used more often
 - GDP deflator
 - Consumer Price Index (CPI)
 - Producer Price Index (PPI)

CPI

- CPI is a measure of the overall cost of the goods and services bought by a typical consumer
- Computing CPI
 - Fix the basket: Determine what items are most important to the typical consumer
 - Find the prices: Find the prices of each of the goods and services in the basket for each point in time
 - Compute the basket's cost
 - Choose a base year and compute the index by dividing the price of the basket in one year by the price in the base year and multiplying by 100

Computing CPI

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2001	\$1	\$2
2002	2	3
2003	3	4

Step 3: Compute the Cost of the Basket of Goods in Each Year

2001	$(\$1 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 2 \text{ hamburgers}) = \8
2002	$(\$2 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$3 \text{ per hamburger} \times 2 \text{ hamburgers}) = \14
2003	$(\$3 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$4 \text{ per hamburger} \times 2 \text{ hamburgers}) = \20

Computing CPI and inflation

Step 4: Choose One Year as a Base Year (2001) and Compute the Consumer Price Index in Each Year

2001	$(\$8/\$8) \times 100 = 100$
2002	$(\$14/\$8) \times 100 = 175$
2003	$(\$20/\$8) \times 100 = 250$

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2002	$(175 - 100)/100 \times 100 = 75\%$
2003	$(250 - 175)/175 \times 100 = 43\%$

The CPI basket (weighing diagram)

base: 2012

Item	Rural	Urban
Food and Beverages	54	36
Pan, tobacco, intoxicants	3	1
Clothing and footwear	7	5.5
Housing		21.5
Fuel and light	8	5.5
Miscell (edun, health, transport)	27	29.5

CPI – the cost of living

- An objective of CPI is to measure the changes in the cost of living
 - How much incomes should rise to maintain a standard of living

Centre likely to hike dearness allowance by 3%

The dearness allowance for employees and pensioners is worked out on the basis of the latest Consumer Price Index for Industrial Workers (CPI-IW) brought out by the Labour Bureau every month

August 06, 2023 11:01 am | Updated 11:01 am IST - New Delhi

CPI – the LTCG

- How do we calculate the long-term capital gains? (useful for the income tax purpose)

INCOME-TAX

NOTIFICATION

New Delhi, the 10th April, 2023

S.O. 1692(E).— In exercise of the powers conferred by clause (v) of the *Explanation* to section 48 of the Income-tax Act, 1961 (43 of 1961), the Central Government hereby makes the following further amendments in the notification of the Government of India, Ministry of Finance (Department of Revenue), Central Board of Direct Taxes, published in the Gazette of India, Extraordinary, *vide* number S.O. 1790(E), dated the 5th June, 2017, namely:-

2. In the said notification, in the TABLE, after serial number 22, the following serial number and entries relating thereto, shall be inserted, namely:-

TABLE

Sl. No.	Financial Year	Cost Inflation Index (provisional)
(1)	(2)	(3)
“23	2023-24	348”.

3. This notification shall come into force with effect from the 1st day of April, 2024 and shall, accordingly, apply in relation to the assessment year 2024-25 and subsequent assessment years.

[Notification No. 21 /2023/F.No.370142/5/2023-TPL]

CPI - measurement issues

- CPI as a measure of cost of living
 - Substitution bias
 - Change in prices over time is not constant across all goods. The consumers will substitute the costly goods with cheaper alternatives, but the basket (weighing diagram) does not change
 - Introduction of new goods
 - Means consumers have a larger basket to choose from and it affects the cost of living
- Considering some of these issues, some countries such as the US, use a chained CPI
 - A chained index “augments the fixed-weight price index with a changing-weight system that accounts for the consumer substitution

PPI

- PPI measures the cost of a basket of goods and services bought by firms (and not the consumers)
- In India, we had used Wholesale Price Index (WPI) to measure inflation for a long period
 - After the seventh base year revision, WPI (2011-12 series) is based on data on about 700 items/ commodities collected from different centers across the country
 - Weight attached to each item is “based on the value of production adjusted for net imports”
 - WPI basket does not cover services

Composition of WPI in India

(only illustration)



The price level in India

Base year: 2011-12

Source: NAS 2021

2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	Item
3	4	5	6	7	8	9	10	11	2
100.0	106.9	112.5	113.9	109.7	111.6	114.9	119.8	121.8	wholesale prices (all commodities)
100.0	111.4	122.4	125.1	124.6	128.9	130.6	134.2	143.3	primary articles (food & non-food articles & minerals)
100.0	105.3	108.5	111.2	109.2	110.7	113.8	117.9	118.3	manufactured products
									consumer prices*
100.0	110.6	121.2	128.7	135.9	142.6	147.8	152.2	158.7	CPI (Rural)
100.0	109.1	119.1	125.9	131.1	136.3	141.2	146.7	154.6	CPI (Urban)
100.0	109.9	120.3	127.4	133.6	139.7	144.7	149.6	156.8	CPI (Combined)
100.0	110.4	121.1	128.7	136.0	141.6	146.0	153.9	165.5	CPI (IW)

Inflation targeting in India, recent changes

- Since 2014, following the recommendations of the Expert Committee to Revise and Strengthen the Monetary Policy Framework (headed by Dr. Urjit Patel), [the RBI has been targeting](#) CPI (combined)
- CPI (combined) is based on a combination of CPI for rural and urban areas and is compiled by MoSPI
- The inflation target of 4 % (in terms of CPI) [was kept for the period](#) 2016-21 and has been retained for the period of next five years, 2021-26. The target is decided jointly by the Government and RBI. There is an upper limit of 6 % and a lower limit of 2 % to this target
- Earlier, the RBI used to target the WPI-based measure. The information on WPI is compiled by the Office of Economic Advisor, Ministry of Commerce and Industry

Real and nominal interest rates

- Nominal interest rate (i)
 - Is the interest rate usually reported without a correction for inflation
 - What is the interest rate we get on the money in the savings bank account?
- Real interest rate (r)
 - Is the interest rate adjusted for the effect of inflation

Most popular movies of all time in the US

1. Avatar: USD 749 million
2. Titanic: USD 601 million
3. The Dark Knight: USD 533 million

These figures are inflation *unadjusted*

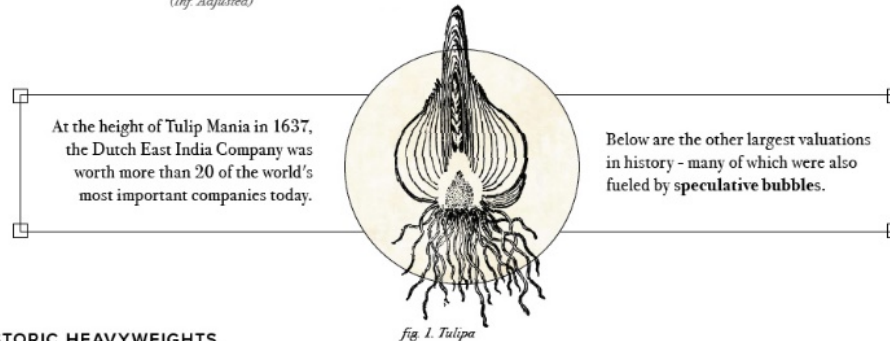
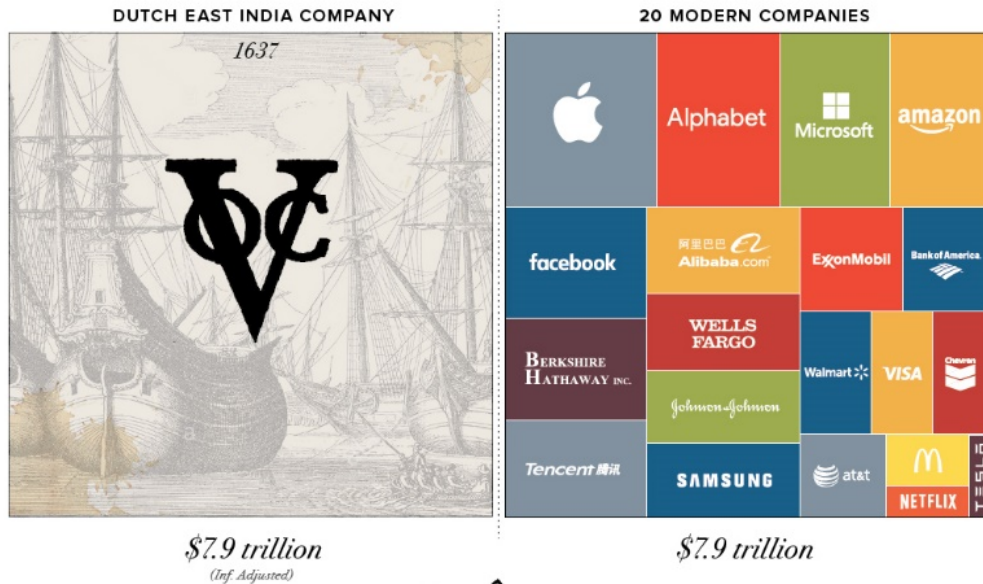
Most popular movies of all time in US

inflation adjusted

Film	Year of Release	Total Domestic Gross (in millions of 2001 dollars)
1. <i>Gone with the Wind</i>	1939	\$1,002
2. <i>Star Wars</i>	1977	866
3. <i>The Sound of Music</i>	1965	695
4. <i>E.T.: The Extra-Terrestrial</i>	1982	687
5. <i>Titanic</i>	1997	640
6. <i>The Ten Commandments</i>	1956	639
7. <i>Jaws</i>	1975	625
8. <i>Doctor Zhivago</i>	1965	591
9. <i>The Jungle Book</i>	1967	519
10. <i>Snow White and the Seven Dwarfs</i>	1937	518

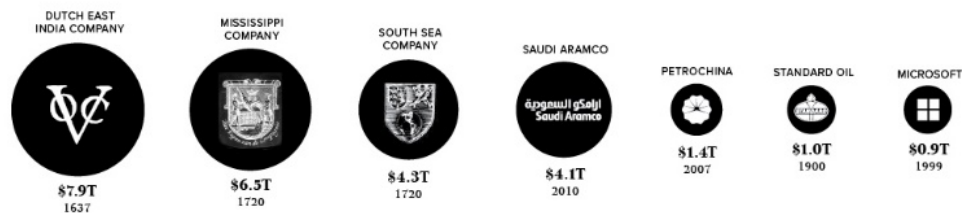
THE MOST VALUABLE COMPANIES OF ALL-TIME

Modern juggernauts like Apple and Amazon don't even come close



HISTORIC HEAVYWEIGHTS

Peak value (adj. for inflation)



SOURCE: The Motley Fool, Barry Ritholtz, Sheridan Titman

visualcapitalist.com



Source: <https://www.visualcapitalist.com/most-valuable-companies-all-time/>
(2017)

References

- Relevant sections in
 - Chapters 23 and 24, Mankiw
 - Chapter 20, Samuelson and Nordhaus
- Those interested in more on India may refer to
 - National Account Statistics and CPI (available at the website of Ministry of Statistics and Programme Implementation)
 - Ministry of Commerce and Industry (for WPI)